

Table of $Z\varphi$ -classes for PerfectGroup(7680, 2) and PerfectGroup(7680, 3)

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GAP 4.15 computations, $2 \leq n \leq 31$

The two perfect groups considered have the same centre $C_4 \times C_2$ and the same faithful action $\text{Out}(P) \simeq D_8 \simeq \text{Aut}(C_4 \times C_2)$ on this centre. Hence the value $N(n)$ below applies separately to each of PerfectGroup(7680, 2) and PerfectGroup(7680, 3). The cumulative count starts at $n = 2$. Times are the `runtimeMs` values reported by the GAP computations, converted to seconds. For $n \geq 16$, the computation uses the `count-only` version, without cumulative storage of class representatives.

n	$ S = 8n$	$ E = 7680n$	$\# Z\varphi$	cumulative	time (s)	cumulative time (s)
2	16	15 360	10	10	0.479	0.479
3	24	23 040	1	11	0.260	0.739
4	32	30 720	83	94	4.568	5.307
5	40	38 400	1	95	0.262	5.569
6	48	46 080	20	115	3.991	9.560
7	56	53 760	1	116	0.488	10.048
8	64	61 440	790	906	69.024	79.072
9	72	69 120	2	908	2.003	81.075
10	80	76 800	20	928	4.356	85.431
11	88	84 480	1	929	0.347	85.778
12	96	92 160	256	1 185	66.457	152.235
13	104	99 840	1	1 186	0.406	152.641
14	112	107 520	20	1 206	11.829	164.470
15	120	115 200	1	1 207	3.250	167.720
16	128	122 880	11 123	12 330	1 378.358	1 546.078
17	136	130 560	1	12 331	4.296	1 550.374
18	144	138 240	50	12 381	21.936	1 572.310
19	152	145 920	1	12 382	0.505	1 572.815
20	160	153 600	269	12 651	54.878	1 627.693
21	168	161 280	2	12 653	2.436	1 630.129
22	176	168 960	20	12 673	3.712	1 633.841
23	184	176 640	1	12 674	0.445	1 634.286
24	192	184 320	3 602	16 276	716.237	2 350.523
25	200	192 000	2	16 278	2.197	2 352.720
26	208	199 680	20	16 298	3.807	2 356.527
27	216	207 360	5	16 303	14.206	2 370.733
28	224	215 040	253	16 556	45.867	2 416.600
29	232	222 720	1	16 557	0.613	2 417.213
30	240	230 400	40	16 597	25.654	2 442.867
31	248	238 080	1	16 598	0.827	2 443.694

Thus, for each of the two perfect residuals,

$$\sum_{n=2}^{31} N(n) = 16\,598.$$

The cumulative time of the computations reported in the table is

$$2\,443.694 \text{ s} \simeq 40.73 \text{ min.}$$

Values of n contributing the largest shares of the total runtime

n	time (s)	share of total time	# $Z\varphi$	share of $Z\varphi$ -classes
16	1 378.358	56.40%	11 123	67.01%
24	716.237	29.31%	3 602	21.70%
8	69.024	2.82%	790	4.76%
12	66.457	2.72%	256	1.54%
Top four combined	2 230.076	91.26%	15 771	95.02%

The four values $n = 16, 24, 8, 12$ thus account for **91.26% of the total computation time** and **95.02% of all $Z\varphi$ -classes** obtained for $2 \leq n \leq 31$. The relation between computational cost and the number of classes is particularly clear for $n = 16$ and $n = 24$: together, these two rows account for 85.71% of the total time and 88.72% of the classes.

The row $n = 16$ alone yields 11 123 classes, while the row $n = 24$ yields 3 602.

Remark on the runtime for $n = 16$. The GAP computation record gives `runtimeMs = 1378358`, i.e. 1 378.358 s. An external measurement obtained from the difference of `Runtime()` values in the session gives about 24.6 minutes; for consistency with the other rows, the table retains the internal `runtimeMs` value.